

This is equivalent to claiming that the augmented matrix $[a_1 a_2 b]$ must have a solution set.

$$\begin{bmatrix} 1 & -2 & 4 \\ 4 & -3 & 1 \\ -2 & 7 & h \end{bmatrix} \xrightarrow[\text{-4} \times \text{Row 1}]{\text{Row 2} = \text{Row 2}} \begin{bmatrix} 1 & -2 & 4 \\ 0 & 5 & -15 \\ -2 & 7 & h \end{bmatrix} \xrightarrow[\text{+2} \times \text{Row 1}]{\text{Row 3} = \text{Row 3}} \begin{bmatrix} 1 & -2 & 4 \\ 0 & 5 & -15 \\ 0 & 3 & 8h \end{bmatrix}$$

$$\xrightarrow[\times 1/5]{\text{Row 2} = \text{Row 2}} \begin{bmatrix} 1 & -2 & 4 \\ 0 & 1 & -3 \\ 0 & 3 & 8h \end{bmatrix} \xrightarrow[\text{-3} \times \text{Row 2}]{\text{Row 3} = \text{Row 3}} \begin{bmatrix} 1 & -2 & 4 \\ 0 & 1 & -3 \\ 0 & 0 & 17h \end{bmatrix}$$

Verification that the final matrix is in echelon form

1. If $17h \neq 0$, then no zero rows in the matrix exist. $\therefore$ The condition that all zero rows must be below all non-zero rows ~~must~~ is vacuously true.

If $17h = 0$, then the 3rd Row is 0. Row 1 and Row 2 are non-zero rows. $\therefore$ The condition that all non-zero rows must be above all zero rows is True.

2. Row 1 Leading entry = 1 in Column 1
 Row 2 Leading entry = 1 in Column 2
 Row 3 Leading entry (possible) = $17h$ in Column 3

$\therefore$ The condition that the leading entry of each row must be in a column to the right of the columns of the leading entries of all rows above it is True.

3. Row 2 Column 1 = Row 3 Column 1 = 0
 Row 3 Column 2 = 0

$\therefore$ The condition that for each pivot column, all elements below the pivot position is 0 is True.

So, the matrix $\begin{bmatrix} 1 & -2 & 4 \\ 0 & 1 & -3 \\ 0 & 0 & 17h \end{bmatrix}$ is in echelon form. $\square$

There exists a row of the form $[0 \ 0 \ 17h]$. By Theorem 2, if $17h = 0$, then the solution set exists. $\therefore h = -17$